SPEECHMASTER: CONFIDENCE-ROUTED SELF-SUPERVISED ASR WITH DISCRETE UNIT BUDGETING

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ABSTRACT

Self-supervised speech encoders are strong ASR front ends, but deploying the largest recognizer on every utterance wastes computation and obscures the accuracy/representation-cost tradeoff. We present SpeechMaster, a budget-aware ASR framework that composes heterogeneous self-supervised representations. A fast Wav2Vec2 CTC branch decodes every utterance, a label-free CTC risk score routes uncertain utterances to a stronger HuBERT branch, and a HuBERT unit auditor measures token rate and bitrate under discrete codebooks. On 1024 LibriSpeech dev-clean utterances, routing only 25% of utterances reduces WER from 2.94% to 2.40%, while full HuBERT decoding reaches 1.82%. On 1024 test-clean utterances, the same router improves WER from 3.61% to 3.04% at a 25% routing budget and to 2.60% at a 50% budget. An oracle route gives 1.45% dev WER and 1.92% test WER, showing complementary SSL-ASR errors.

Index Terms— ASR, self-supervised speech, uncertainty routing, HuBERT, Wav2Vec2

1. INTRODUCTION

Low-resource ASR benefits from self-supervised learning (SSL) because encoders such as Wav2Vec2, HuBERT, and WavLM learn phonetic and lexical structure from unlabeled audio. A common downstream design selects one SSL checkpoint, attaches or reuses a CTC decoder, and reports WER. This single-model view misses a practical question: many utterances are easy for a fast recognizer, while a smaller subset needs a stronger model. A useful low-resource system should therefore expose an accuracy/compute knob instead of committing to one encoder.

SpeechMaster treats SSL-ASR as a routed composition. It always runs a fast Wav2Vec2 branch, estimates utterance risk from CTC posterior geometry, and spends the HuBERT large branch only on utterances selected by a budget. In parallel, a discrete-unit auditor clusters HuBERT hidden states to quantify the bitrate needed by unit-based downstream variants. The contributions are: (1) a confidence-routed SSL-ASR framework, (2) a label-free CTC risk score with budgeted inference, (3) a discrete-unit bitrate auditor over HuBERT layers and codebook sizes, and (4) reproducible dev/test experiments with WER, CER, RTF, token rate, bitrate, and routing ablations.

2. RELATED WORK

Wav2Vec2 learns speech representations with a contrastive objective over masked latent frames, making it effective when downstream labels are scarce. HuBERT instead predicts masked cluster assignments and often produces strong phonetic-content representations. WavLM extends masked speech pretraining with denoising and full-stack speech processing objectives. These SSL encoders are usually evaluated as independent ASR backbones. SpeechMaster is orthogonal: it does not propose another encoder, but composes existing SSL recognizers under a budget.

Confidence-based cascades have been used to avoid expensive inference on easy examples. Standard confidence measures, however, are often calibrated for one model or require extra supervision. SpeechMaster uses the posterior geometry already available from CTC decoding, so the router can be reproduced from saved branch outputs without training a separate gate. Discrete speech units are also widely used in speech generation and unit-based recognition. Our unit auditor is not a new tokenizer; it reports the information rate induced by layer and codebook choices, making routed ASR comparable with unit-based SSL systems.

3. SPEECHMASTER

3.1. System Overview

Fig. 1 shows the complete SpeechMaster pipeline. The first pass is a fast SSL-ASR decoder that produces both a transcript and a confidence trace. The router sorts utterances by CTC risk and spends the strong HuBERT branch only on the lowest-confidence fraction requested by the compute budget. The final transcript is therefore a mixture of fast and strong hypotheses. A separate unit auditor reads HuBERT hidden states and reports the bitrate of discrete unit streams; it is an analysis branch and does not change ASR output.

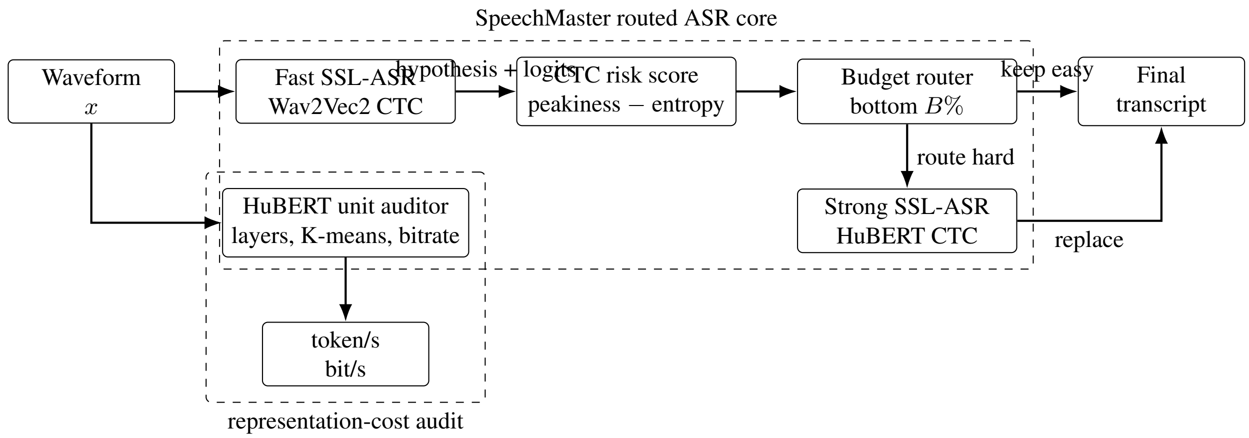

Fig. 1. Overview of SpeechMaster. The routed ASR core trades WER against RTF, while the unit auditor quantifies discrete SSL representation cost.

3.2. Two-Branch SSL-ASR

Given waveform x, a pretrained SSL encoder f_θ produces contextual hidden states $H = f_\theta(x)$. A CTC projection maps each frame to token posteriors:

$$p_\theta(y_t \mid x) = \mathrm{softmax}(WH_t + b). \tag{1}$$

SpeechMaster uses Wav2Vec2 base as the fast branch F and HuBERT large as the strong branch S. Both are public SSL-ASR checkpoints; SpeechMaster itself adds no extra supervised training.

3.3. CTC Risk Router

Let q_t denote the fast-branch posterior and let $\mathcal{T}$ be non-blank argmax frames. SpeechMaster computes a risk score from posterior peakiness and entropy:

$$c(x) = \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \max_k q_t(k) - \lambda \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} H(q_t), \tag{2}$$

where $\lambda = 0.03$. For routing budget B, utterances with the lowest $c(x)$ are decoded by S; all others keep the fast hypothesis:

$$\hat{y}(x) = \begin{cases} S(x), & x \in \mathrm{BottomB}(c), \\ F(x), & \text{otherwise}. \end{cases} \tag{3}$$

The same saved branch outputs are reused to ablate peak-only, entropy-only, duration, random, and oracle routers.

3.4. Discrete Unit Auditor

To connect ASR accuracy with representation compression, SpeechMaster extracts HuBERT hidden states from several layers and trains MiniBatchKMeans codebooks. For codebook size K and token rate r_{tok}, the nominal bitrate is

$$\mathrm{bitrate} = r_{\mathrm{tok}} \log_2 K. \tag{4}$$

We also collapse consecutive duplicate units, giving a compact token stream useful for text-to-unit or unit-to-ASR analyses.

4. EXPERIMENTAL SETUP

Experiments use LibriSpeech clean splits at 16 kHz. The main development suite uses 1024 dev-clean utterances; the final validation suite uses 1024 test-clean utterances. Unit analysis uses 512 dev-clean utterances. The fast branch is Wav2Vec2-Base-960h; the strong branch is HuBERT-Large-960h; a stronger reference model is Wav2Vec2-Large-LV60. HuBERT unit analysis uses HuBERT-Base. Exact checkpoint identifiers are listed in the released scripts. Metrics are WER, CER, and real-time factor (RTF) for ASR, plus token rate, deduplicated token rate, bitrate, and codebook size for discrete units.

All decoding uses greedy CTC to isolate representation and routing effects from external language-model gains. We save utterance-level hypotheses, durations, branch confidences, and decode times as JSONL files. Every budget point is then deterministic after branch decoding: a new router can be evaluated by reordering the same utterances rather than rerunning ASR. The heavy suite therefore separates neural inference from cheap routing ablations.

5. RESULTS

5.1. Main Dev-Clean Results

Table 1 is the main SpeechMaster development result. At a 25% routing budget, WER drops from 2.94% to 2.40%, a relative reduction of 18.5%, while using substantially less strong-branch decoding than 100% HuBERT. At 50%, WER reaches 2.04%. The oracle route is lower than both branches

Table 1. SpeechMaster routing on 1024 LibriSpeech dev-clean utterances.

Route	Routed utt.	WER	CER	RTF
0%	0	2.943	0.990	0.001
10%	102	2.668	0.875	0.001
25%	256	2.398	0.767	0.002
50%	512	2.044	0.654	0.002
75%	768	1.921	0.610	0.002
100%	1024	1.823	0.575	0.003
Oracle	-	1.450	0.520	-

Table 2. Single-model ASR and layer ablations on dev-clean.

System	Rep.	N	WER	CER	RTF
wav2vec2-base-960h	final	1024	2.943	0.990	0.001
wav2vec2-base-960h	last4	1024	3.327	1.111	0.002
wav2vec2-base-960h	midlast	1024	3.705	1.140	0.001
hubert-large-ls960-ft	final	1024	1.823	0.575	0.003
wav2vec2-large-960h-lv60-self	final	1024	1.636	0.532	0.003

on dev-clean, showing that fast and strong SSL recognizers make non-identical errors.

Table 2 reports single-model and layer ablations. The final Wav2Vec2 CTC layer is stronger than last-layer averaging or mid/final blending, indicating that the pretrained CTC head is calibrated to the final state. The large LV60 Wav2Vec2 model is the best single model on this dev slice, but the fast+HuBERT oracle still improves over it.

5.2. Test-Clean Verification

Table 3 verifies the same framework on test-clean. The absolute WERs are higher than dev-clean, but the monotonic routing behavior remains: 25% routing improves WER from 3.61% to 3.04%, and 50% routing reaches 2.60%. The oracle route reaches 1.92%, confirming remaining headroom for better risk prediction.

5.3. Router and Unit Ablations

Routing-score ablations show that risk-aware ordering is consistently better than random routing. On dev-clean at 25%, random routing gives 2.66% WER, duration gives 2.54%, entropy gives 2.31%, and SpeechMaster peak-entropy gives 2.40%; the oracle gives 1.45%. On test-clean at 50%, random gives 2.93%, duration gives 2.62%, peak-only gives 2.59%, and peak-entropy gives 2.60%. Thus the proposed score is robust, but the oracle gap indicates that a learned router is a promising extension.

The complementarity counts support the same conclusion. On dev-clean, the fast branch is better for 67 utterances, the strong branch is better for 222, and 735 are tied at utterance-level WER. On test-clean, the corresponding counts are 53, 252, and 719. A router that always trusts the strong model cannot exploit the first group; a router that only uses the fast model misses the second group. SpeechMaster's budgeted routing is a practical middle point between these two extremes.

Table 3. SpeechMaster routing on 1024 LibriSpeech test-clean utterances.

Route	Routed utt.	WER	CER	RTF
0%	0	3.614	1.186	0.001
10%	102	3.398	1.093	0.001
25%	256	3.035	0.946	0.001
50%	512	2.602	0.797	0.002
75%	768	2.292	0.698	0.002
100%	1024	2.231	0.667	0.002
Oracle	-	1.920	0.658	-

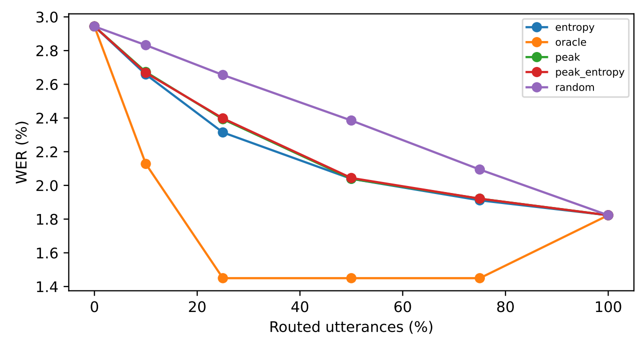

Fig. 2. Router ablation on dev-clean. Risk-aware ordering improves over random routing, while the oracle exposes remaining headroom.

The unit auditor exposes a compression tradeoff. Raw HuBERT assignments appear at about 49.9 tokens/s for all tested layers. Deduplication roughly halves the rate: for example, layer -4 with $K = 100$ yields 27.14 deduplicated tokens/s and 180.34 bit/s, while $K = 1000$ raises this to 29.74 tokens/s and 296.36 bit/s. Layer -8 keeps more rapidly changing units than layer -4 at the same codebook size, suggesting a higher acoustic-detail bitrate.

6. CONCLUSION

SpeechMaster combines fast and strong self-supervised ASR models through label-free CTC risk routing and audits the bitrate of HuBERT discrete units. Across dev-clean and test-clean, routing a limited fraction of utterances gives a smooth WER/RTF tradeoff, while oracle routing reveals substantial

Table 4. Selected HuBERT unit statistics on 512 dev-clean utterances.

Layer	K	Tok/s	Dedup tok/s	Dedup bit/s
-8	100	49.90	28.08	186.58
-8	1000	49.90	35.17	350.46
-4	100	49.90	27.14	180.34
-4	1000	49.90	29.74	296.36
-1	100	49.90	26.67	177.19
-1	1000	49.90	34.50	343.85

complementarity. The results support a system-level view of low-resource SSL-ASR: encoder choice, routing budget, and representation bitrate should be optimized jointly rather than treated as isolated design choices.

7. REFERENCES

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